

SMARTATHON'26

Department of Artificial Intelligence & Data Science
Dhanalakshmi Srinivasan University × ONEYEs
24-Hour National-Level Hackathon | 09–10 September 2026
Confirmed Problem Statements

SMART CITIES

1. AI-Assisted Land Document Verification System

Problem Statement

Land registration in Tamil Nadu is already fully online through TNREGINET — patta, chitta, encumbrance certificate, and sale deed can all be submitted digitally. But once submitted, the actual checking of these documents against each other is still done manually by an officer, which is where mismatches and fraud slip through.

Expected Solution

Build a system that takes a set of submitted land documents and checks them against each other for inconsistencies — mismatched survey numbers, ownership mismatches, missing details, duplicate applications — and flags exactly which documents disagree and why, before the case reaches a human officer for final approval.

2. Chennai Flood Access Risk Mapper

Problem Statement

During heavy rainfall in Chennai, some roads become unsafe or impassable, but there is no single system that tells people or officials in advance which roads are actually at risk before the water rises — decisions are currently made reactively, road by road, as reports come in.

Expected Solution

Build a map-based system that uses rainfall, elevation, drainage, and road network data to identify which road segments are likely to become high-risk during heavy rain, rank them by expected disruption, and suggest safer alternate routes. It must work even for rainfall patterns it has not seen before, not just roads that have flooded historically.

3. Citizen Grievance & Government Order Auto-Routing System

Problem Statement

When a citizen files a complaint, or a District Collector issues an order, it has to reach the right government department to be acted on. Right now this routing is done manually, which is slow, and a single complaint often actually contains more than one issue that belongs to different departments.

Expected Solution

Build a system that reads a complaint or order, works out which department(s) it actually belongs to, pulls out any deadline or required action, and routes it correctly — including splitting a single complaint that covers multiple issues across the right departments, instead of tagging the whole thing under one department.

SMART AUTOMATION

4. Food Factory Process Control System

Problem Statement

In a food factory, the process goes buy → clean → check → chop/grind → pack → sell. Each stage currently runs on its own, so a problem introduced early — bad raw material, a missed cleaning step — often isn't caught until much later, by which point the work already done on that batch is wasted.

Expected Solution

Build a centralized system that tracks every stage of a batch's journey — what came in, cleaning status, quality check result, processing progress, packing, and sale — giving one live view of the whole process. It should also use the early-stage data to predict, before the batch reaches packing, whether it is likely to have a quality problem later — without ever being told in advance which past batches failed and why.

5. Sudden Absence Task Reassignment System

Problem Statement

When a worker doesn't show up for their shift, someone has to quickly decide who else can take over their task. Right now this is a rushed manual call, and it often pulls the wrong person or causes delay.

Expected Solution

Build a system that, the moment an employee is marked absent, automatically works out who should take over their task — based on who is actually capable of it and who has room in their own workload — without leaving that person's own task short-staffed in the process. It should work out who's capable of what from past task-performance history, since no clean skill chart exists to look up.

7. Development of Indigenous Contactless Integrated Ship Monitoring Systems (ISMS)

Problem Statement

Ships require regular inspection to identify structural damage, corrosion, deformation, and operational abnormalities. Current inspection methods often involve manual checks and physical access, making the process time consuming and difficult in hazardous or inaccessible areas.

Expected Solution

Build an indigenous contactless system that uses cameras, LiDAR, thermal, acoustic, or other remote sensing technologies to detect and record structural and operational abnormalities in real time, reducing manual inspection and enabling early fault detection.

8. AI-Powered Autonomous Warehouse Monitoring and Inventory Automation System

Problem Statement

Warehouses require continuous monitoring of inventory, storage locations, product movement, and worker activity. Manual inventory verification and tracking can result in misplaced products, incorrect stock records, inefficient picking routes, and delayed identification of damaged goods.

Expected Solution

Build an intelligent warehouse automation system capable of continuously understanding inventory movement and automatically detecting operational problems. The system should use cameras, RFID, barcode/QR scanning, IoT sensors, or other suitable technologies to track products from receiving to storage, picking, packing, and dispatch. It should maintain real-time inventory records, identify misplaced or missing items, detect damaged goods and unusual movement, optimize picking and movement routes, monitor storage conditions, and provide alerts or recommendations when operational problems are detected.

The system should support an autonomous dashboard that gives warehouse managers a live view of inventory status, product locations, movement history, worker activity, and detected anomalies.

6. AI Scholarship & Government Scheme Eligibility Matcher

Problem Statement

Tamil Nadu runs 15+ overlapping scholarship and welfare schemes with different eligibility rules based on community, income, gender, disability, and course — and many students who genuinely qualify miss out simply because checking every scheme manually is overwhelming. Several schemes also cannot be claimed together.

Expected Solution

Build a system that takes a student's profile and matches it against multiple scheme eligibility rules, then recommends the single best-fit scheme or valid combination — not just a list of every scheme they technically qualify for, since some of those combinations aren't actually allowed together.